Identifying Regional Bottlenecks in Global Trade

GLOBAL LOGISTICS ANALYSIS 2026

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Problem Definition

Which economies (regions/countries) are experiencing the highest median port stay times for specific vessel types?

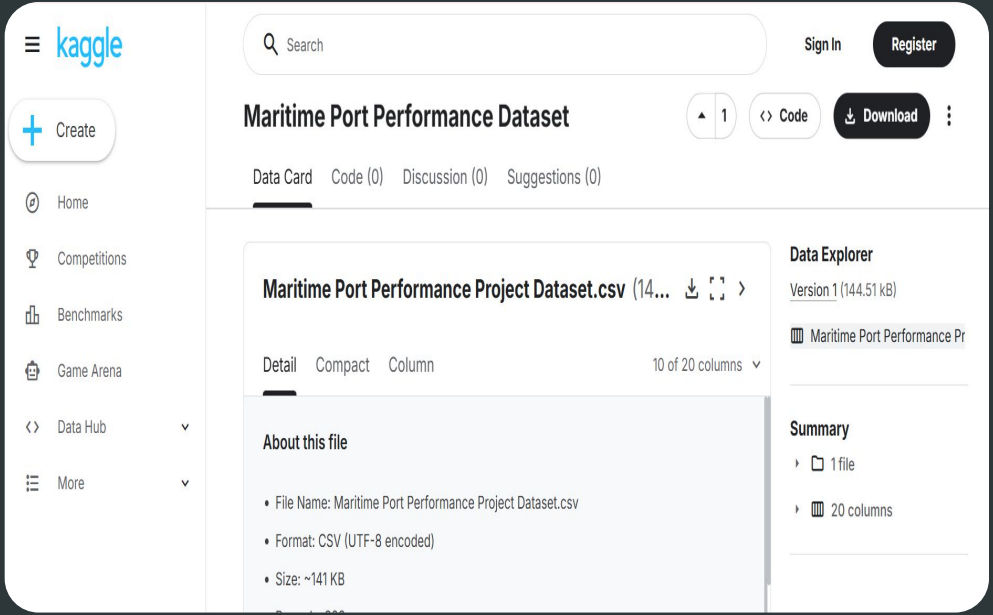
Comparison Example

Container ships vs. Bulk carriers

Extract

Data extracted from **Kaggle.com**

Downloaded in a **.csv format**



Transform

Initial Data Assessment

Identified that messy raw data lacked actionable insights or accurate values.

Renaming & Localization

Standardized "Economy Level" by reclassifying into Regions for better clarity.

Temporal Splitting

Segmented "**Period**" into separate **Year** and **Semester** fields for better filtering.`

0 Missing Value Imputation

Replaced nulls with 0 to maintain integrity without deleting records.

Final Export

Converted processed Data Frame into a structured format using the `.to_csv()` method.

Load

Table Generation
The cleaned table with required headings were generated.

Selected Columns
Only required heads: `Semester, Year, Period_Code, Max_Cargo_DWT, Avg_Size_GT, Ship_Type, Region`

Data Export
Downloaded in .csv format for further downstream analysis.

Output File
File name: **`Cleaned_Maritime_Data.csv`**

Exploratory Data Analysis (EDA)

Scope Isolation

Filtered the dataset to exclude global aggregates, ensuring regional performance comparisons remain unskewed and accurate.

Hierarchical Grouping

Condensed granular maritime records into a concise matrix by aggregating mean port times across Economic Region and Ship Type.

Dimensional Pivoting

Reshaped the data into a spatial-numeric structure, mapping vessel categories against geographical dimensions.

Normalization & Mapping

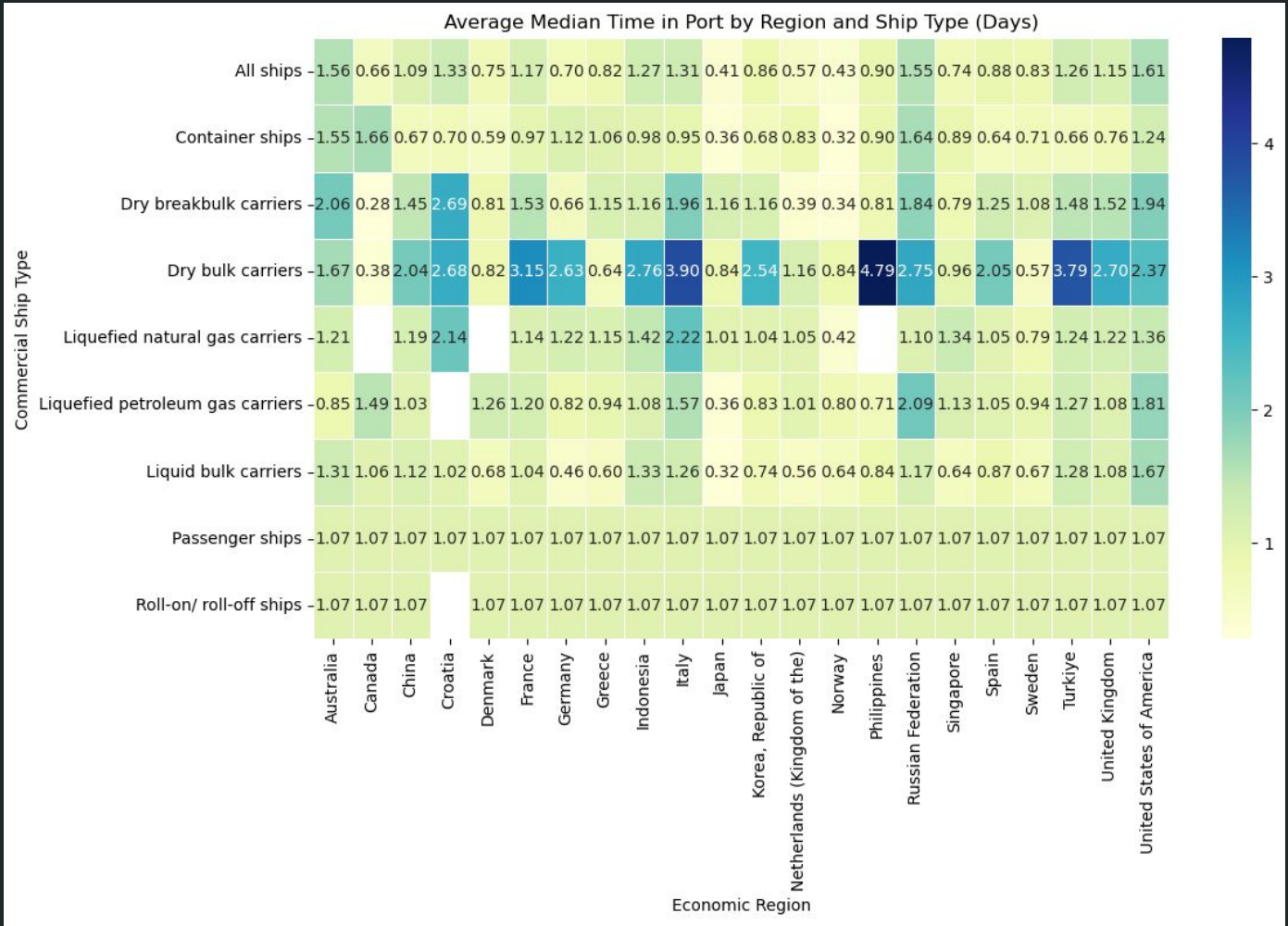
Translated raw efficiency benchmarks into a gradient scale to prepare for visual interpretation.

Bottleneck Identification

Deployed a Heatmap to convert complex matrices into an intuitive, color-coded overview.

Decision Support

Facilitated evidence-based allocation by highlighting historical congestion patterns.


Average Median Time in Port by Region and Ship Type (Days)
Commercial Ship Type
All ships 1.56 0.66 1.09 1.33 0.75 1.17 0.70 0.82 1.27 1.31 0.41 0.86 0.57 0.43 0.90 1.55 0.74 0.88 0.83 1.26 1.15 1.61
Container ships 1.55 1.66 0.67 0.70 0.59 0.97 1.12 1.06 0.98 0.95 0.36 0.68 0.83 0.32 0.90 1.64 0.89 0.64 0.71 0.66 0.76 1.24
Dry breakbulk carriers 2.06 0.28 1.45 2.69 0.81 1.53 0.66 1.15 1.16 1.96 1.16 1.16 0.39 0.34 0.81 1.84 0.79 1.25 1.08 1.48 1.52 1.94
Dry bulk carriers 1.67 0.38 2.04 2.68 0.82 3.15 2.63 0.64 2.76 3.90 0.84 2.54 1.16 0.84 4.79 2.75 0.96 2.05 0.57 3.79 2.70 2.37
Liquefied natural gas carriers 1.21 1.19 2.14 1.14 1.22 1.15 1.42 2.22 1.01 1.04 1.05 0.42 1.10 1.34 1.05 0.79 1.24 1.22 1.36
Liquefied petroleum gas carriers 0.85 1.49 1.03 1.26 1.20 0.82 0.94 1.08 1.57 0.36 0.83 1.01 0.80 0.71 2.09 1.13 1.05 0.94 1.27 1.08 1.81
Liquid bulk carriers 1.31 1.06 1.12 1.02 0.68 1.04 0.46 0.60 1.33 1.26 0.32 0.74 0.56 0.64 0.84 1.17 0.64 0.87 0.67 1.28 1.08 1.67
Passenger ships 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07
Roll-on/ roll-off ships 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07 1.07
Australia
Canada
China
Croatia
Denmark
France
Germany
Greece
Indonesia
Italy
Japan
Korea, Republic of
Netherlands (Kingdom of the)
Norway
Philippines
Russian Federation
Singapore
Spain
Sweden
Turkiye
United Kingdom
United States of America
Economic Region
4
3
2
1

Machine Learning (ML)

Workspace Initialization

Established a dedicated environment and ingested the refined maritime dataset for predictive modeling.

Variable Partitioning

Defined the feature set (X) and the target variable (y) to establish the model's objective functions.

Preprocessing Orchestration

Constructed a transformation pipeline to automate data scaling and encoding for consistent input.

Pipeline Integration

Unified the preprocessing steps with the machine learning algorithm into a single, cohesive model pipeline.

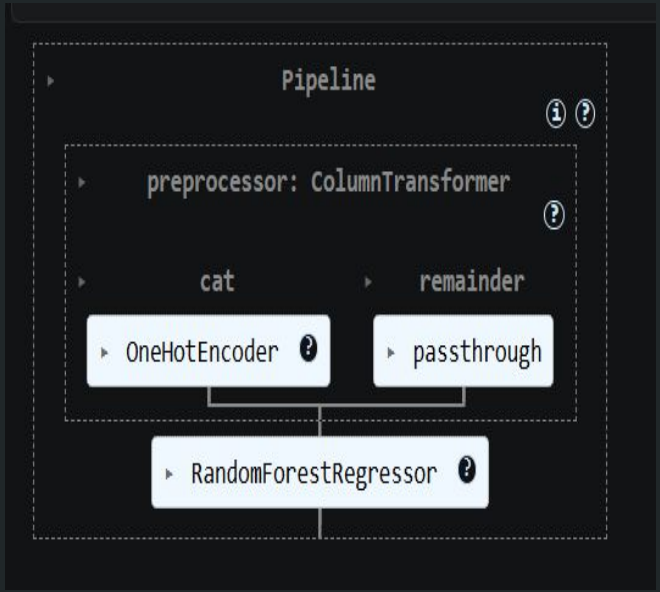
Validation & Training

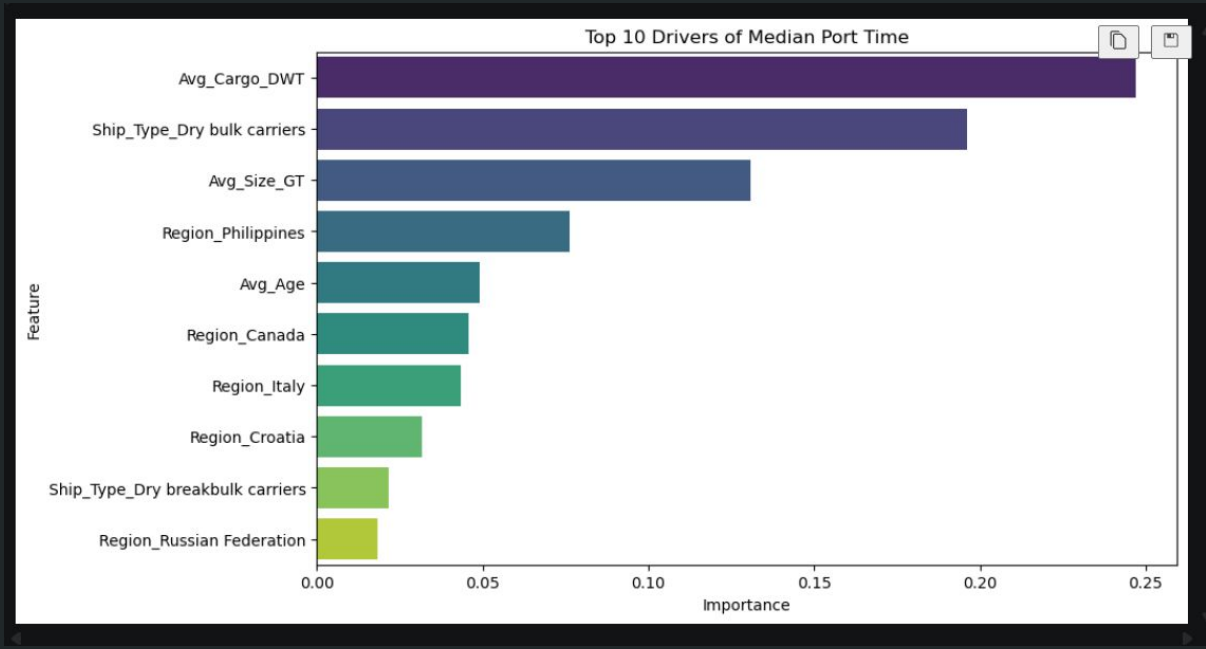
Partitioned the data into training and testing subsets, executing the model fit on the training data.

Predictive Evaluation

Assessed model accuracy by generating predictions on the hold-out set and calculating performance metrics.

Interpretability Analysis: Visualized Feature Importance to rank the variables most critical to predicting maritime port efficiency.


Pipeline
preprocessor: ColumnTransformer
cat
remainder
OneHotEncoder
passthrough
RandomForestRegressor


Top 10 Drivers of Median Port Time
Avg_Cargo_DWT
Ship_Type_Dry bulk carriers
Avg_Size_GT
Region_Philippines
Avg_Age
Region_Canada
Region_Italy
Region_Croatia
Ship_Type_Dry breakbulk carriers
Region_Russian Federation
Feature
0.00
0.05
0.10
0.15
0.20
0.25
Importance

Statistical Analysis

Descriptive Baselining

Defined "normal" port performance using mean and standard deviation to isolate outliers from consistent trends.

Correlation Mapping

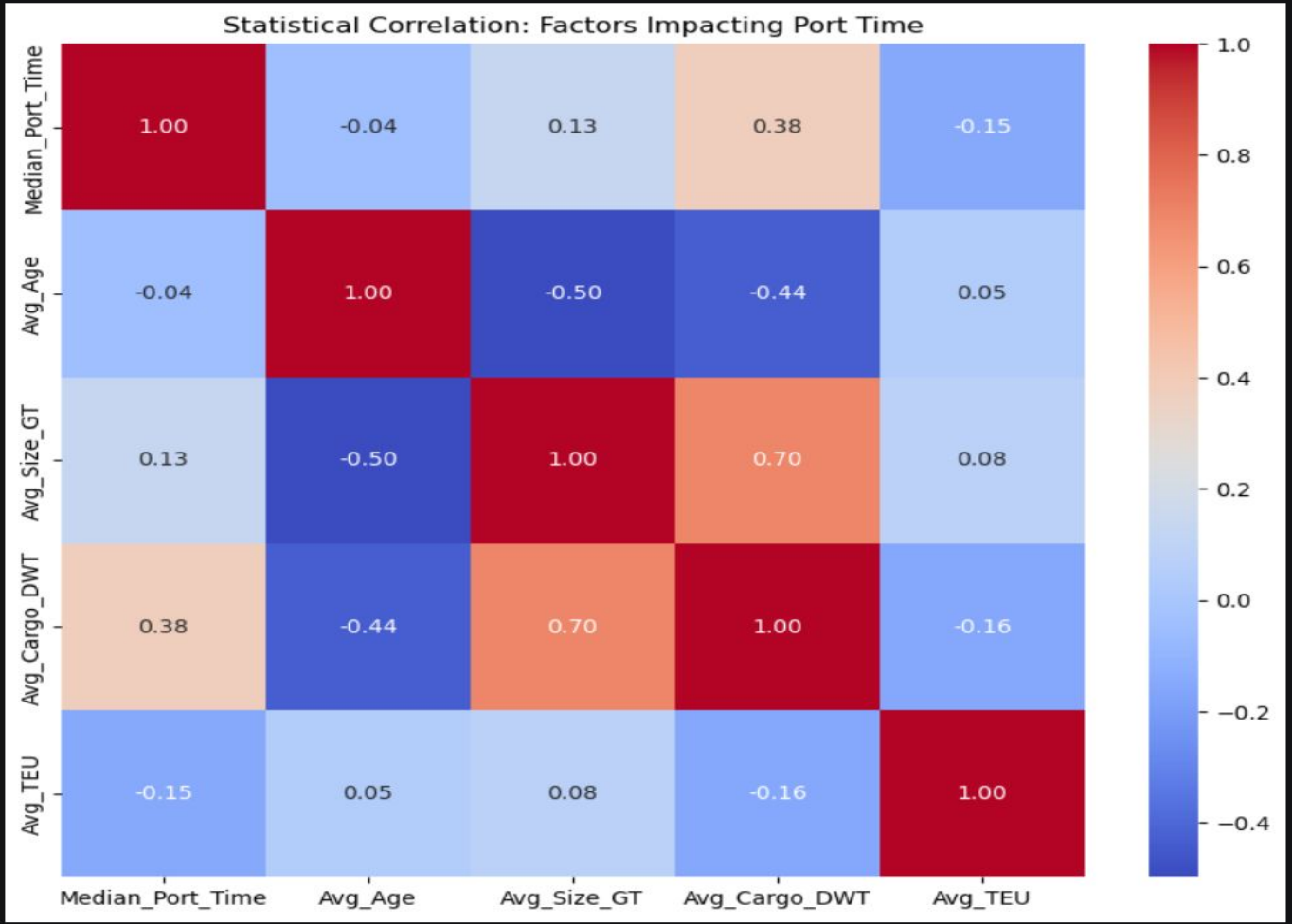
Quantified the strength of relationships between vessel attributes and latency to identify true operational drivers.

Significance Testing

Applied T-tests and P-values to mathematically prove that efficiency gaps are statistically significant, not random.

Empirical Validation

Established a rigorous, evidence-based foundation to justify strategic recommendations and academic conclusions.


Statistical Correlation: Factors Impacting Port Time
Median_Port_Time
Avg_Age
Avg_Size_GT
Avg_Cargo_DWT
Avg_TEU
1.00
-0.04
0.13
0.38
-0.15
-0.04
1.00
-0.50
-0.44
0.05
0.13
-0.50
1.00
0.70
0.08
0.38
-0.44
0.70
1.00
-0.16
-0.15
0.05
0.08
-0.16
1.00
Median_Port_Time
Avg_Age
Avg_Size_GT
Avg_Cargo_DWT
Avg_TEU
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

Project Analytical Framework

EDA & Heatmaps

Visually maps port stays against regions and vessel types to pinpoint high-latency "ground truths" and operational bottlenecks.

Machine Learning

Analyzes patterns in vessel age, size, and type to determine if delays are caused by regional inefficiency or vessel-specific characteristics.

Feature Importance

Ranks variables influencing port latency, allowing decision-makers to prioritize specific policy or infrastructure investments.

Statistical Validation

Uses T-tests and P-values to verify that observed performance gaps are statistically significant structural dependencies.

Strategic Goal: Driving evidence-based infrastructure and policy optimization for global maritime trade efficiency.

Project Impact: Outputs & Strategic Value

Core Deliverables (The Outputs)

Predictive Dashboard
ML-based tool forecasting vessel stay times based on age, size, and type.

Dynamic Heatmap
Spatial map identifying critical congestion bottlenecks by region.

Feature Priority & Benchmarks
Ranked analysis of critical variables and statistically proven performance KPIs.

Strategic Insights & Implementation

Root Cause & Optimization
Distinguishes management issues and enables proactive resource scheduling.

Investment Roadmap
Provides the business case for targeted upgrades like dredging and automation.

Policy Support
Empowers leadership with empirical evidence for SLAs and policy reforms.

Bottom Line: Transforms raw maritime records into a defensible, evidence-based strategy for enhancing global supply chain resilience.

Conclusion

Integrated Pipeline

Successfully unified data engineering, ML modeling, and statistical validation into one cohesive framework.

Actionable Insights

Transformed raw maritime data into predictive tools that identify and explain structural port bottlenecks.

Empirical Authority

Provided a mathematically rigorous foundation to justify strategic operational and policy recommendations.

Strategic Value

Delivered a scalable system to optimize vessel turnaround times and enhance global supply chain resilience.

Summary: The project delivers a defensible, data-driven framework for modernizing maritime logistics and global trade infrastructure.

Thanks!
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